

## Parameter and Statistic - Answers

### 1. Online Orders

**Parameter:** The mean amount spent on all online orders placed last month,  $\mu$ . Its value is unknown.

**Statistic:** The mean amount spent on the 380 sampled orders,  $\bar{x} = \$74.60$ .

### 2. Left-Handed Students

**Parameter:** The proportion of all students at the college who are left-handed,  $p$ . Its value is unknown.

**Statistic:** The proportion of the 240 sampled students who were left-handed:

$$\hat{p} = \frac{29}{240} \approx 0.121 \quad (12.1\%)$$

### 3. Jacket Production

**Parameter:** The mean sewing time for all 1,200 jackets made on Monday,  $\mu = 41.8$  minutes.

**Statistic:** The mean sewing time for the 60 sampled jackets,  $\bar{x} = 43.1$  minutes.

### 4. Mobile Payments

**Parameter:** The proportion of all customers who pay with the store's mobile app,  $p$ . Its value is unknown.

**Statistic:** The proportion of the 800 observed customers who paid with the app:

$$\hat{p} = \frac{436}{800} = 0.545 \quad (54.5\%)$$

### 5. Defective Buttons

**Parameter:** The proportion of all 5,000 buttons in the shipment that were defective:

$$p = \frac{125}{5000} = 0.025 \quad (2.5\%)$$

**Statistic:** The proportion of the 200 sampled buttons that were defective:

$$\hat{p} = \frac{8}{200} = 0.04 \quad (4\%)$$